

NICOLETTE MASHABA

Junior Software Engineer | AI & Full-Stack Development

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Location: South Africa (Open to Remote, Hybrid, On-site)

LinkedIn: linkedin.com/in/nicolette-mashaba

GitHub: github.com/NickiMash17

Portfolio: nicmash-portfolio.vercel.app

Professional Summary

Junior Software Engineer with hands-on experience building, deploying, and maintaining AI-powered and full-stack web applications. Strong foundation in Python, JavaScript/TypeScript, REST APIs, and frontend-backend integration. Microsoft Azure certified (AZ-204) with practical experience deploying applications to Azure, Vercel, Render, and Netlify. Known for strong problem-solving skills, secure development practices, and the ability to translate real-world problems into working software solutions.

Technical Skills

Languages: Python, JavaScript, TypeScript, C#, SQL

Frontend: React, HTML5, CSS3, Tailwind CSS, Responsive Design

Backend: Node.js (Express), Python (FastAPI), ASP.NET Core, REST APIs

AI & Automation: OpenAI API, LLM Integration, Prompt Engineering, Structured Outputs

Databases: MongoDB, SQL Server, PostgreSQL, Firebase

Cloud & Deployment: Microsoft Azure (AZ-204), Vercel, Render, Netlify, Docker basics

Tools & Practices: Git, GitHub, .gitignore, Environment Variables, CI/CD concepts, Agile/Scrum

Experience

Technical Trainer (Volunteer) — TechBridle Foundation

2026 – Present

- Mentor learners in web development fundamentals, JavaScript, React, and backend concepts.
- Support debugging and problem-solving; guide Git workflows and secure project setup practices.

Software Engineering Intern — Coast IT (Pty) Ltd

2025

- Supported development of application features within an Agile team environment.
- Assisted with testing, bug fixing, and code reviews to improve application stability and quality.

AI Model Trainer — Outlier AI

2024

- Evaluated AI-generated outputs and improved response quality through structured feedback.
- Performed prompt testing and model behaviour analysis to improve clarity and reliability.

Projects

AIMS Compliance Interrogator (Hackathon Winner)

GitHub: github.com/NickiMash17/AIMS-Firefly

- Identified inefficiencies in manual compliance review and built an AI-powered tool to analyse structured inputs and flag risks.
- Designed controlled LLM prompt workflows to ensure consistent, explainable outputs instead of free-form responses.
- Solved challenges around sensitive data handling by enforcing backend-only AI calls and secure environment configuration.
- Deployed the full-stack application on Microsoft Azure; project won a hackathon and was featured on an Australian technology podcast.

Empower AI (Live Application)

Live: <https://empower-ai-gamma.vercel.app>

- Addressed the challenge of delivering personalised guidance at scale by building an AI-powered well-being assistant.
- Improved response usefulness by refining prompts and constraining outputs to structured, actionable guidance.
- Implemented secure frontend–backend integration; project is in active development with implementation details kept private.

Interview Replay (Live Application)

Live: <https://interviewreplay.netlify.app>

- Solved the problem of limited interview practice by creating an AI-driven mock interview and feedback tool.
- Iteratively improved AI feedback quality by structuring responses into clear strengths and improvement points.
- Deployed the application live and maintained production configuration using environment variables and clean Git practices.

BookReviewApp — Full-Stack Web Application

GitHub: github.com/NickiMash17/BookReviewApp

- Built a full-stack book review platform to manage and moderate user-generated content.
- Designed RESTful APIs using ASP.NET Core and Entity Framework Core to handle reviews, users, and data persistence.
- Addressed challenges related to data integrity and application structure by applying clean architecture principles.
- Implemented authentication, CRUD workflows, and database migrations to support a scalable backend system.

Certifications

Microsoft Certified: Azure Developer Associate (AZ-204)

Microsoft Certified: Azure Data Fundamentals (DP-900)

J.P. Morgan Software Engineering Virtual Experience

Education

Occupational Certificate: Software Engineering (NQF Level 6)

Expected 2026

CTU Training Solutions, South Africa

Information Technology: Programming Foundations (NQF Level 4)

2024

CTU Training Solutions, South Africa

Additional Experience

Junior Navigation Officer — South African Navy

2015 – 2021

- Developed discipline, teamwork, and attention to detail in high-responsibility environments.

Availability

Available immediately for full-time opportunities.